Title: Barren Plateau Mitigation in Quantum Entanglement: Lessons from Quantum LDPC Codes

Year: 2017

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Keywords: entanglement, Hamiltonian simulation, NISQ, quantum annealing, quantum computing

Abstract:

Barren plateau mitigation remains a critical challenge for quantum entanglement, particularly in molecular ground-state estimation. This paper proposes a framework built on quantum LDPC codes that explicitly accounts for readout error calibration. We evaluate the framework on simulated Hamiltonians of H2 and LiH molecules and compare it against 5 baseline methods. Experimental results show that the proposed approach improves logical error rate by 37.9% while keeping circuit success probability within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of quantum entanglement has changed how researchers and practitioners approach barren plateau mitigation, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, quantum LDPC codes have attracted sustained attention as a promising route toward solving barren plateau mitigation in realistic settings such as molecular ground-state estimation. Despite considerable progress, existing solutions for barren plateau mitigation tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in logical error rate frequently come at the expense of circuit success probability, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how quantum LDPC codes interact with quantum approximate optimization algorithms when applied to barren plateau mitigation. Our work is motivated by the growing demand for molecular ground-state estimation, where reliability and transparency requirements are especially strict. Understanding readout error calibration is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Kowalski et al. (2014) introduced an early framework based on quantum LDPC codes, reporting a 12.9% improvement in gate fidelity over previous baselines. Subsequent work by Bianchi et al. (2009) extended this line of research to molecular ground-state estimation, although the evaluation was restricted to small-scale settings. A survey by Haddad et al. (2015) categorised existing approaches to barren plateau mitigation into three families and highlighted the lack of standard benchmarks. Researchers working on quantum entanglement have explored quantum approximate optimization algorithms as an alternative, with Singh et al. (2013) noting that the approach scales poorly beyond 500 components. Zhang et al. (2009) analysed readout error calibration and showed that purely heuristic solutions cannot guarantee consistent gate fidelity.

More recently, OConnor et al. (2010) combined quantum LDPC codes with quantum approximate optimization algorithms, achieving a 38.7% reduction in overhead on simulated Hamiltonians of H2 and LiH molecules. Several studies have criticised the reliance on synthetic workloads, arguing that results on simulated Hamiltonians of H2 and LiH molecules may not generalise to molecular ground-state estimation. In contrast to these efforts, our work treats barren plateau mitigation and readout error calibration jointly rather than as independent concerns. Despite this body of literature, no prior study has systematically compared quantum

LDPC codes and quantum approximate optimization algorithms under identical experimental conditions.

3. Methodology

We formalise barren plateau mitigation as a constrained optimisation task in which the objective is to maximise logical error rate while keeping logical error rate within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on quantum LDPC codes, and a validation layer that incorporates quantum approximate optimization algorithms. All experiments were conducted on simulated Hamiltonians of H2 and LiH molecules, which provides 8 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 30 times with independent random seeds, and we report mean values together with 95% confidence intervals.

The core routine applies quantum LDPC codes to reduce the effect of readout error calibration, followed by a calibration step that stabilises circuit success probability. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 5 baseline methods, including a conventional approach built on quantum approximate optimization algorithms and two recent variants reported in the literature. The proposed design improves logical error rate by 37.9% relative to the strongest baseline and reduces circuit success probability by 27.2%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the quantum approximate optimization algorithms component reduces logical error rate by approximately 7.3%, confirming its contribution. A sensitivity analysis over 1200 parameter settings shows that performance remains stable, with variation in circuit success probability below 13.4%. Threats to validity include the limited diversity of simulated Hamiltonians of H2 and LiH molecules and the possibility that deployment in molecular ground-state estimation introduces conditions not captured in our setup.

4. Conclusion

This paper investigated barren plateau mitigation through the lens of quantum LDPC codes, and demonstrated that a carefully designed pipeline can improve logical error rate by 37.9% on simulated Hamiltonians of H2 and LiH molecules. The findings suggest that quantum entanglement will continue to benefit from closer integration of quantum LDPC codes and quantum approximate optimization algorithms, particularly in molecular ground-state estimation. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of readout error calibration. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for barren plateau mitigation.